

```

#import libraries
import numpy as np
import pandas as pd

#Load dataset
df=pd.read_csv("/content/1775656337232-titanic_dataset.csv")
df

{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 891,\n  \"fields\": [\n    {\n      \"column\": \"PassengerId\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 257,\n        \"min\": 1,\n        \"max\": 891,\n        \"num_unique_values\": 891,\n        \"samples\": [\n          710,\n          440,\n          841\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Survived\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 0,\n        \"max\": 1,\n        \"num_unique_values\": 2,\n        \"samples\": [\n          1,\n          0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Pclass\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 1,\n        \"max\": 3,\n        \"num_unique_values\": 3,\n        \"samples\": [\n          3,\n          1\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Name\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 891,\n        \"samples\": [\n          \"Moubarek, Master. Halim Gonios (\\\"William George\\\")\",\n          \"Kvillner, Mr. Johan Henrik Johannesson\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Sex\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"female\",\n          \"male\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Age\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 14.526497332334044,\n        \"min\": 0.42,\n        \"max\": 80.0,\n        \"num_unique_values\": 88,\n        \"samples\": [\n          0.75,\n          22.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"SibSp\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1,\n        \"min\": 0,\n        \"max\": 8,\n        \"num_unique_values\": 7,\n        \"samples\": [\n          1,\n          0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Parch\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 0,\n        \"max\": 6,\n        \"num_unique_values\": 7,\n        \"samples\": [\n          0,\n          1\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\":

```

```

\"Ticket\",\\n      \\\"properties\\\": {\\n      \\\"dtype\\\": \\\"string\\\",\\n
\\\"num_unique_values\\\": 681,\\n      \\\"samples\\\": [\\n
\\\"11774\\\",\\n      \\\"248740\\\"\\n      ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n
n      },\\n      {\\n      \\\"column\\\": \\\"Fare\\\",\\n      \\\"properties\\\": {\\n
\\\"dtype\\\": \\\"number\\\",\\n      \\\"std\\\": 49.693428597180905,\\n
\\\"min\\\": 0.0,\\n      \\\"max\\\": 512.3292,\\n
\\\"num_unique_values\\\": 248,\\n      \\\"samples\\\": [\\n
11.2417,\\n      51.8625\\n      ],\\n      \\\"semantic_type\\\":
\\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n      },\\n      {\\n
\\\"column\\\": \\\"Cabin\\\",\\n      \\\"properties\\\": {\\n      \\\"dtype\\\":
\\\"category\\\",\\n      \\\"num_unique_values\\\": 147,\\n
\\\"samples\\\": [\\n      \\\"D45\\\",\\n      \\\"B49\\\"\\n      ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n
n      },\\n      {\\n      \\\"column\\\": \\\"Embarked\\\",\\n      \\\"properties\\\":
{\\n      \\\"dtype\\\": \\\"category\\\",\\n      \\\"num_unique_values\\\":
3,\\n      \\\"samples\\\": [\\n      \\\"S\\\",\\n      \\\"C\\\"\\n
],\\n      \\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n
}\\n      }\\n      ]\\n      }\", \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

```
df.head()#display first 5 rows
```

```

{ \"summary\": \"{\\n  \\\"name\\\": \\\"df\\\",\\n  \\\"rows\\\": 891,\\n  \\\"fields\\\": [\\n
n    {\\n      \\\"column\\\": \\\"PassengerId\\\",\\n      \\\"properties\\\": {\\n
\\\"dtype\\\": \\\"number\\\",\\n      \\\"std\\\": 257,\\n      \\\"min\\\": 1,\\n
\\\"max\\\": 891,\\n      \\\"num_unique_values\\\": 891,\\n
\\\"samples\\\": [\\n      710,\\n      440,\\n      841\\n
],\\n      \\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n
}\\n    },\\n    {\\n      \\\"column\\\": \\\"Survived\\\",\\n
\\\"properties\\\": {\\n      \\\"dtype\\\": \\\"number\\\",\\n      \\\"std\\\":
0,\\n      \\\"min\\\": 0,\\n      \\\"max\\\": 1,\\n
\\\"num_unique_values\\\": 2,\\n      \\\"samples\\\": [\\n      1,\\n
0\\n      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n      }\\n    },\\n    {\\n      \\\"column\\\":
\\\"Pclass\\\",\\n      \\\"properties\\\": {\\n      \\\"dtype\\\": \\\"number\\\",\\n
\\\"std\\\": 0,\\n      \\\"min\\\": 1,\\n      \\\"max\\\": 3,\\n
\\\"num_unique_values\\\": 3,\\n      \\\"samples\\\": [\\n      3,\\n
1\\n      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n      }\\n    },\\n    {\\n      \\\"column\\\":
\\\"Name\\\",\\n      \\\"properties\\\": {\\n      \\\"dtype\\\": \\\"string\\\",\\n
\\\"num_unique_values\\\": 891,\\n      \\\"samples\\\": [\\n
\\\"Moubarek, Master. Halim Gonios (\\\\\\\\\\\"William George\\\\\\\\\\\")\\\",\\n
\\\"Kvillner, Mr. Johan Henrik Johannesson\\\"\\n      ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n
n    },\\n    {\\n      \\\"column\\\": \\\"Sex\\\",\\n      \\\"properties\\\": {\\n
\\\"dtype\\\": \\\"category\\\",\\n      \\\"num_unique_values\\\": 2,\\n
\\\"samples\\\": [\\n      \\\"female\\\",\\n      \\\"male\\\"\\n      ],\\n
n      \\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n
}\\n    },\\n    {\\n      \\\"column\\\": \\\"Age\\\",\\n      \\\"properties\\\": {\\n
n      \\\"dtype\\\": \\\"number\\\",\\n      \\\"std\\\": 14.526497332334044,\\n

```

```

n      \ "min\": 0.42,\n      \ "max\": 80.0,\n
\ "num_unique_values\": 88,\n      \ "samples\": [\n      0.75,\n
22.0\n      ],\n      \ "semantic_type\": \ "\",\n
\ "description\": \ "\n      }\n      },\n      {\n      \ "column\":
\ "SibSp\",\n      \ "properties\": {\n      \ "dtype\": \ "number\",\n
\ "std\": 1,\n      \ "min\": 0,\n      \ "max\": 8,\n
\ "num_unique_values\": 7,\n      \ "samples\": [\n      1,\n
0\n      ],\n      \ "semantic_type\": \ "\",\n
\ "description\": \ "\n      }\n      },\n      {\n      \ "column\":
\ "Parch\",\n      \ "properties\": {\n      \ "dtype\": \ "number\",\n
\ "std\": 0,\n      \ "min\": 0,\n      \ "max\": 6,\n
\ "num_unique_values\": 7,\n      \ "samples\": [\n      0,\n
1\n      ],\n      \ "semantic_type\": \ "\",\n
\ "description\": \ "\n      }\n      },\n      {\n      \ "column\":
\ "Ticket\",\n      \ "properties\": {\n      \ "dtype\": \ "string\",\n
\ "num_unique_values\": 681,\n      \ "samples\": [\n
\ "11774\",\n      \ "248740\",""\n      ],\n
\ "semantic_type\": \ "\",\n      \ "description\": \ "\n      }\n
n      },\n      {\n      \ "column\": \ "Fare\",\n      \ "properties\": {\n
\ "dtype\": \ "number\",\n      \ "std\": 49.693428597180905,\n
\ "min\": 0.0,\n      \ "max\": 512.3292,\n
\ "num_unique_values\": 248,\n      \ "samples\": [\n
11.2417,\n      51.8625\n      ],\n      \ "semantic_type\":
\ "\",\n      \ "description\": \ "\n      }\n      },\n      {\n
\ "column\": \ "Cabin\",\n      \ "properties\": {\n      \ "dtype\":
\ "category\",\n      \ "num_unique_values\": 147,\n
\ "samples\": [\n      \ "D45\",\n      \ "B49\",""\n      ],\n
\ "semantic_type\": \ "\",\n      \ "description\": \ "\n      }\n
n      },\n      {\n      \ "column\": \ "Embarked\",\n      \ "properties\":
{\n      \ "dtype\": \ "category\",\n      \ "num_unique_values\":
3,\n      \ "samples\": [\n      \ "S\",\n      \ "C\",""\n
      ],\n      \ "semantic_type\": \ "\",\n      \ "description\": \ "\n
      }\n      }\n      ]\n      }", "type": "dataframe", "variable_name": "df"}

```

#Remove all the null values

```
df.isnull().sum()#check
```

```

PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64

```

```
df=df.dropna()#remove  
df
```

```
{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 183,\n  \"fields\": [\n    {\n      \"column\": \"PassengerId\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 247,\n        \"min\": 2,\n        \"max\": 890,\n        \"num_unique_values\": 183,\n        \"samples\": [\n          119,\n          252,\n          743\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Survived\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 0,\n        \"max\": 1,\n        \"num_unique_values\": 2,\n        \"samples\": [\n          0,\n          1\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Pclass\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 1,\n        \"max\": 3,\n        \"num_unique_values\": 3,\n        \"samples\": [\n          1,\n          3\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Name\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 183,\n        \"samples\": [\n          \"Baxter, Mr. Quigg Edmond\",\n          \"Strom, Mrs. Wilhelm (Elna Matilda Persson)\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Sex\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"male\",\n          \"female\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Age\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 15.643865966849717,\n        \"min\": 0.92,\n        \"max\": 80.0,\n        \"num_unique_values\": 63,\n        \"samples\": [\n          11.0,\n          6.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"SibSp\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 0,\n        \"max\": 3,\n        \"num_unique_values\": 4,\n        \"samples\": [\n          0,\n          2\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Parch\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 0,\n        \"max\": 4,\n        \"num_unique_values\": 4,\n        \"samples\": [\n          1,\n          4\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Ticket\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 127,\n        \"samples\": [\n          \"PC17595\",\n          \"113806\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Fare\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 53.901052667440937,\n        \"min\": 5.44,\n        \"max\": 512.329,\n        \"num_unique_values\": 146,\n        \"samples\": [\n          5.44,\n          51.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ]\n}}
```

```

\"number\", \n          \"std\": 76.34784270040574, \n          \"min\": 0.0, \n          \"max\": 512.3292, \n          \"num_unique_values\": 93, \n          \"samples\": [\n              29.7, \n              66.6 \n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\" \n      }, \n      {\n          \"column\": \"Cabin\", \n          \"properties\": {\n              \"dtype\": \"string\", \n              \"num_unique_values\": 133, \n              \"samples\": [\n                  \"C7\", \n                  \"C104\" \n              ], \n              \"semantic_type\": \"\", \n              \"description\": \"\" \n          }, \n          {\n              \"column\": \"Embarked\", \n              \"properties\": {\n                  \"dtype\": \"category\", \n                  \"num_unique_values\": 3, \n                  \"samples\": [\n                      \"C\", \n                      \"S\" \n                  ], \n                  \"semantic_type\": \"\", \n                  \"description\": \"\" \n              } \n          } \n      ], \n      \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

#Remove unwanted columns

```

df=df.drop(columns=['Cabin','Ticket','Name'], errors='ignore')
df

```

```

{\"summary\": \"{ \n  \"name\": \"df\", \n  \"rows\": 183, \n  \"fields\": [\n    {\n      \"column\": \"PassengerId\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 247, \n        \"min\": 2, \n        \"max\": 890, \n        \"num_unique_values\": 183, \n        \"samples\": [\n            119, \n            252, \n            743 \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      }, \n      {\n          \"column\": \"Survived\", \n          \"properties\": {\n              \"dtype\": \"number\", \n              \"std\": 0, \n              \"min\": 0, \n              \"max\": 1, \n              \"num_unique_values\": 2, \n              \"samples\": [\n                  0, \n                  1 \n              ], \n              \"semantic_type\": \"\", \n              \"description\": \"\" \n          }, \n          {\n              \"column\": \"Pclass\", \n              \"properties\": {\n                  \"dtype\": \"number\", \n                  \"std\": 0, \n                  \"min\": 1, \n                  \"max\": 3, \n                  \"num_unique_values\": 3, \n                  \"samples\": [\n                      1, \n                      3 \n                  ], \n                  \"semantic_type\": \"\", \n                  \"description\": \"\" \n              }, \n              {\n                  \"column\": \"Sex\", \n                  \"properties\": {\n                      \"dtype\": \"category\", \n                      \"num_unique_values\": 2, \n                      \"samples\": [\n                          \"male\", \n                          \"female\" \n                      ], \n                      \"semantic_type\": \"\", \n                      \"description\": \"\" \n                  }, \n                  {\n                      \"column\": \"Age\", \n                      \"properties\": {\n                          \"dtype\": \"number\", \n                          \"std\": 15.643865966849717, \n                          \"min\": 0.92, \n                          \"max\": 80.0, \n                          \"num_unique_values\": 63, \n                          \"samples\": [\n                              11.0, \n                              6.0 \n                          ], \n                          \"semantic_type\": \"\", \n                          \"description\": \"\" \n                      }, \n                      {\n                          \"column\": \"SibSp\", \n                          \"properties\": {\n                              \"dtype\": \"number\", \n                              \"std\": 0, \n                              \"min\": 0, \n                              \"max\": 3, \n                              \"num_unique_values\": 4, \n                              \"samples\": [\n                                  0, \n                                  2 \n                              ], \n                              \"semantic_type\": \"\", \n                              \"description\": \"\" \n                          }, \n                          {\n                              \"column\":

```



```

63,\n          \"samples\": [\n          11.0,\n          6.0\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n          },\n          {\n          \"column\":\n          \"SibSp\",\n          \"properties\": {\n          \"dtype\": \"number\",\n          \"std\": 0,\n          \"min\": 0,\n          \"max\": 3,\n          \"num_unique_values\": 4,\n          \"samples\": [\n          0,\n          2\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n          },\n          {\n          \"column\":\n          \"Parch\",\n          \"properties\": {\n          \"dtype\": \"number\",\n          \"std\": 0,\n          \"min\": 0,\n          \"max\": 4,\n          \"num_unique_values\": 4,\n          \"samples\": [\n          1,\n          4\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n          },\n          {\n          \"column\":\n          \"Fare\",\n          \"properties\": {\n          \"dtype\": \"number\",\n          \"std\": 76.34784270040574,\n          \"min\": 0.0,\n          \"max\": 512.3292,\n          \"num_unique_values\": 93,\n          \"samples\": [\n          29.7,\n          66.6\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n          }\n          },\n          {\n          \"column\": \"Embarked\",\n          \"properties\": {\n          \"dtype\": \"category\",\n          \"num_unique_values\": 3,\n          \"samples\": [\n          \"C\",\n          \"S\"\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n          }\n          }\n          ],\n          \"type\": \"dataframe\"}

```

```

import matplotlib.pyplot as plt
import seaborn as sns

```

```

#find overall survival rate

```

```

survival_=df['Survived'].mean()*100
survival_

```

```

np.float64(67.21311475409836)

```

```

#survival rate by gender

```

```

gen_survival=df.groupby('Sex')['Survived'].mean()*100
gen_survival

```

```

Sex
female    93.181818
male      43.157895
Name: Survived, dtype: float64

```

```

#survival rate by age

```

```

age_survival=df.groupby('Age')['Survived'].mean()*100
age_survival

```

```

Age
0.92    100.000000
1.00    100.000000
2.00     33.333333
3.00    100.000000

```

```
4.00      100.000000
```

```
64.00      0.000000
```

```
65.00      0.000000
```

```
70.00      0.000000
```

```
71.00      0.000000
```

```
80.00      100.000000
```

```
Name: Survived, Length: 63, dtype: float64
```

```
#survival rate by Passenger ID
```

```
pass_survival=df[['PassengerId','Survived']]
```

```
pass_survival.head()
```

```
{"summary":{"\n  \"name\": \"pass_survival\",\n  \"rows\": 183,\n  \"fields\": [\n    {\n      \"column\": \"PassengerId\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 247,\n        \"min\": 2,\n        \"max\": 890,\n        \"num_unique_values\": 183,\n        \"samples\": [\n          119,\n          252,\n          743\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Survived\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 0,\n        \"max\": 1,\n        \"num_unique_values\": 2,\n        \"samples\": [\n          0,\n          1\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ],\n  \"type\": \"dataframe\", \"variable_name\": \"pass_survival\"}
```

```
#count total males and females
```

```
count_=df['Sex'].value_counts()
```

```
count_
```

```
Sex
```

```
male      95
```

```
female    88
```

```
Name: count, dtype: int64
```

```
#cross-validation of male and female data
```

```
c_v_=pd.crosstab(df['Sex'],df['Survived'])
```

```
c_v_
```

```
{"summary":{"\n  \"name\": \"c_v_\",\n  \"rows\": 2,\n  \"fields\": [\n    {\n      \"column\": \"Sex\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"male\",\n          \"female\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": 0,\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 33,\n        \"min\": 6,\n        \"max\": 54,\n        \"num_unique_values\": 2,\n        \"samples\": [\n          54,\n          6\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": 1,\n      \"properties\": {\n        \"dtype\": \"number\",
```



```

{"number": 28, "min": 41, "max": 82, "num_unique_values": 2, "samples": [41, 82], "semantic_type": "", "description": "", "type": "dataframe", "variable_name": "c_v"}

```

#correlation b/n passenger class and survival

```
corre=df[['Pclass', 'Survived']].corr()
```

```
corre_
```

```

{"summary": {"name": "corre_", "rows": 2, "fields": [{"column": "Pclass", "properties": {"dtype": "number", "std": 0.7315317991629043, "min": -0.034542191683370464, "max": 1.0, "num_unique_values": 2, "samples": [0.034542191683370464, 1.0], "semantic_type": "", "description": ""}, {"column": "Survived", "properties": {"dtype": "number", "std": 0.7315317991629043, "min": -0.034542191683370464, "max": 1.0, "num_unique_values": 2, "samples": [1.0, -0.034542191683370464], "semantic_type": "", "description": ""}}], "type": "dataframe", "variable_name": "corre_"}

```